

6 Digit Nixie Clock "64Bit" - Driver Board

64Bit comes from the fact that this clock is driven with shift registers. Between the 6 nixie digits, and indicators, 64 bits are sent from the microcontroller to the shift registers to create the display output

This is an STM32F446 based project.

The clock time keeping is handled via a Real Time Clock (RTC) communicated with via I2C. The STM32 also has an on-board RTC that can be utilized. Both RTCs have a battery backup via a replaceable 2032 coin cell.

The interface with the clock is via a rotary encoder with a push button. Interfacing with remote capacitive buttons is possible via connectors on the bottom of the Driver Board.

There is an ESP32-C6 module driving a full-color LCD screen. The ESP32 features wifi and bluetooth allowing for automatic time setting and smart home integration. Communication with the STM32 is via UART.

This clock was designed with safety in mind. There is a dedicated safety circuit that cuts the input and output connections of the high voltage power supply via relays. There are connectors for interfacing external spring pin connections to the cover of the clock, ensuring the high voltage power supply is not on while the lid is off. The safety circuit features dual channels and feedback to the microcontroller. With the spring pin feature, it will provide isolation from the high voltage, even in an event where the microcontroller experiences an error.

The power design of the clock relays on simple components. There are two inputs available to power the clock. A USB-C port and a 12V barrel jack. A boost converter utilizing XL6009 converts (boost) the incoming voltage to 12VDC. Then an LDO creates 5V then another LDO, in series, creates 3.3V. The high voltage power supply typically is set to output 170VDC to 180VDC. It is supplied by a power supply module that sources 12V power.

This driver board interface with the display board PCB via right angle board-to-board connectors. Mounted to the back of the display board is an LED board to provide RGB backlighting to each IN-12 Nixie Tubes.

The circuit boards will be mounted to a 3D printed base via brass standoffs. Overtop of the PCBs will be a glass top. The glass top will have sections where copper tape is applied. Spring pins contact the copper tape to pass connections for the safety circuit.

STM32 MCU

I2C

Interface

Safety Circuit

Power

Connectors

Connectors 2

Kadin Whiting

Sheet: /

File: 64 Bit - Main Board.kicad_sch

Title: Project Overview

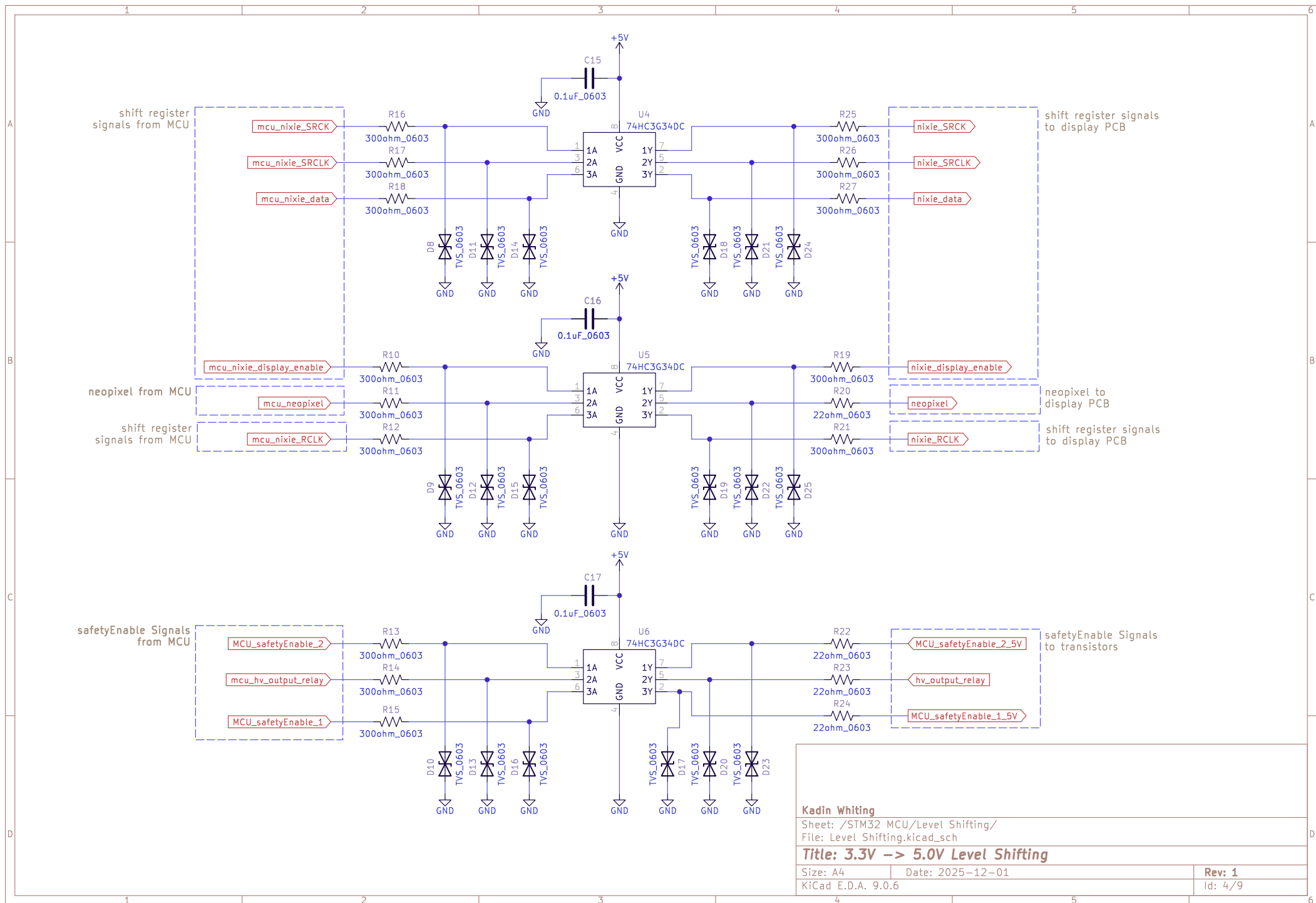
Size: A4

Date: 2025-12-01

Rev: 1

KiCad E.D.A. 9.0.6

Id: 1/9



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Sheet: /STM32 MCU/Level Shifting/
File: Level Shifting.kicad_sch

Title: 3.3V -> 5.0V Level Shifting

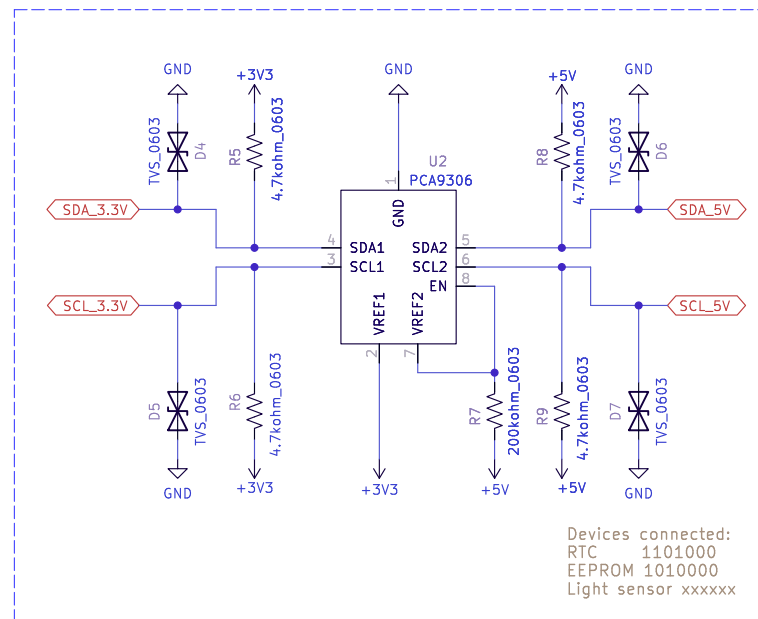
Size: A4 Date: 2025-12-01

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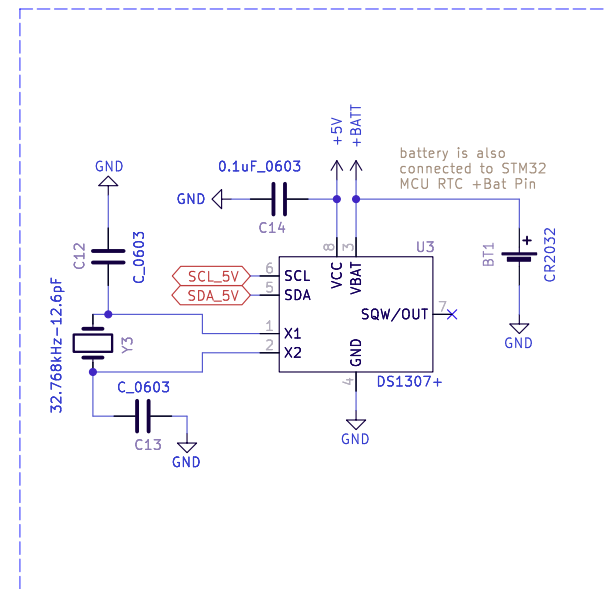
Rev: 1

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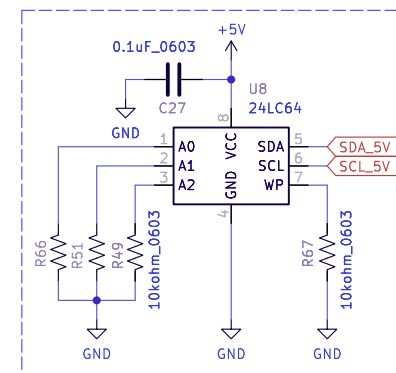
3.3V to 5V I2C level shifter



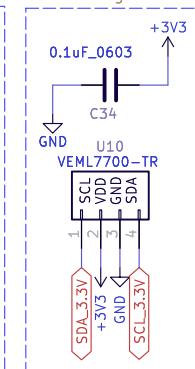
Real Time Clock



EEPROM



ambient light sensor



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Sheet: /I2C/
 File: I2C.kicad_sch

Title: RTC & RTC Level Shifting

Size: A4 Date: 2025-12-01

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Rev: 1

Id: 3/9

right-angle encoder 24 pulses/rotation
integrated momentary button
low pass filters included

encoder_A
encoder_B
encoder_button

+3V3
10kOhm_0603
R29
4.7kOhm_0603
R30
10kOhm_0603
R31
4.7kOhm_0603
R32
10kOhm_0603
R33

10kOhm_0603
R28
0.1uF_0603
C18
10kOhm_0603
R26
0.01uF_0603
C20
10kOhm_0603
R27
0.01uF_0603
C21
0.01uF_0603
C22

D28
TVS_0603
D26
TVS_0603
D27
TVS_0603

U7
CCW
WIPER
C1
CW
NO_1
NO_2
MH1
MH2

EN12-VS20AF20

GND

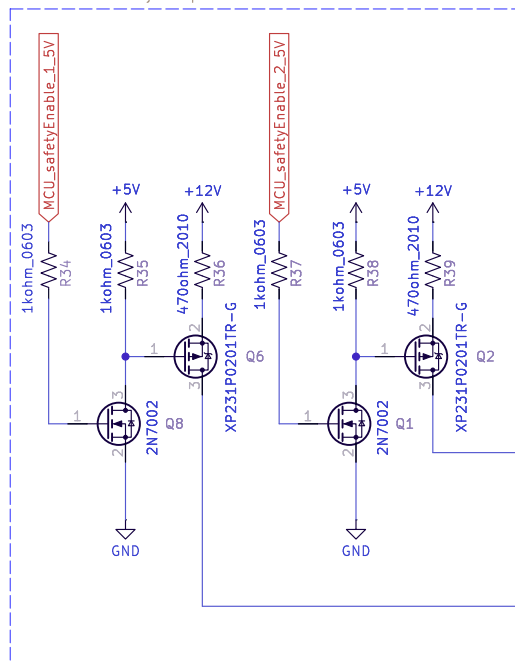
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Size: A4	Date: 2025-12-01
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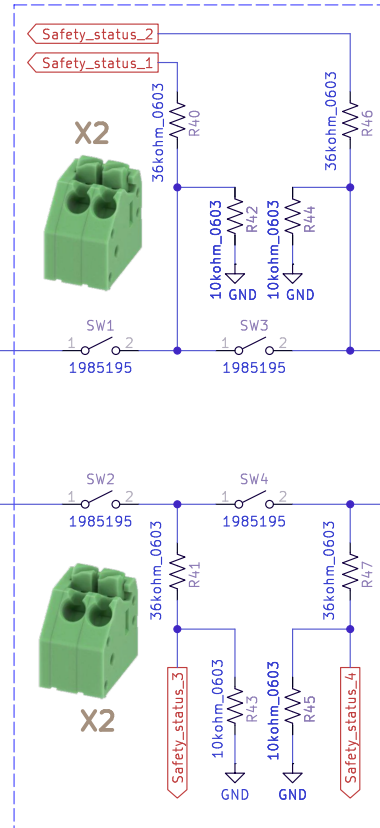
Rev: 1

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two outputs from MCU
to enable safety loop



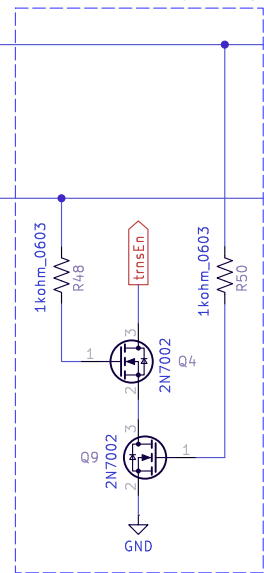
safety contacts for sensing if the lid is installed
feedback to MCU for each contact



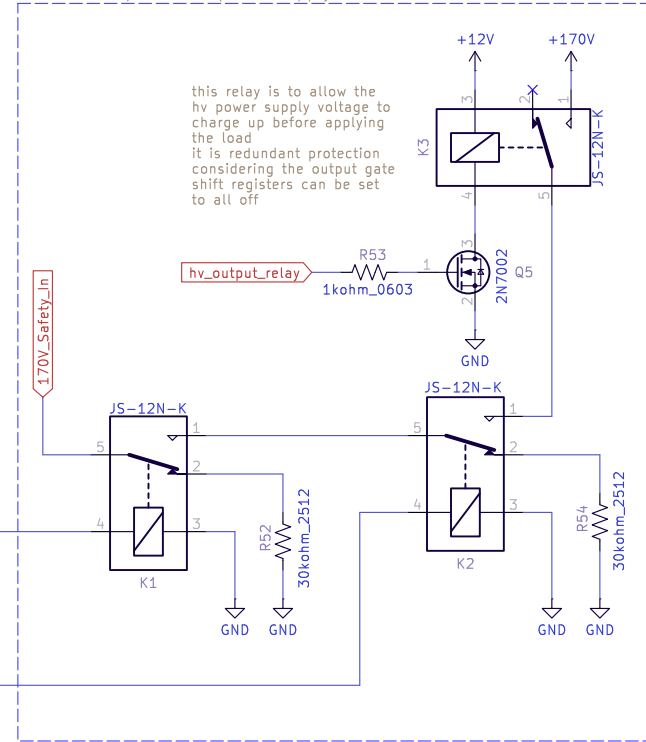
loop through
display PCB



enables HV PSU input relay



controls output of HV power supply



this relay is to allow the
hv power supply voltage to
charge up before applying
the load
it is redundant protection
considering the output gate
shift registers can be set
to all off

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Sheet: /Safety Circuit/

File: Safety_Circuit.kicad_sch

Title: Safety Circuit – High Voltage PSU Control

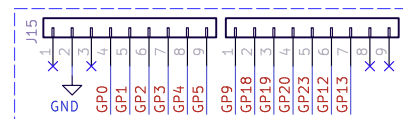
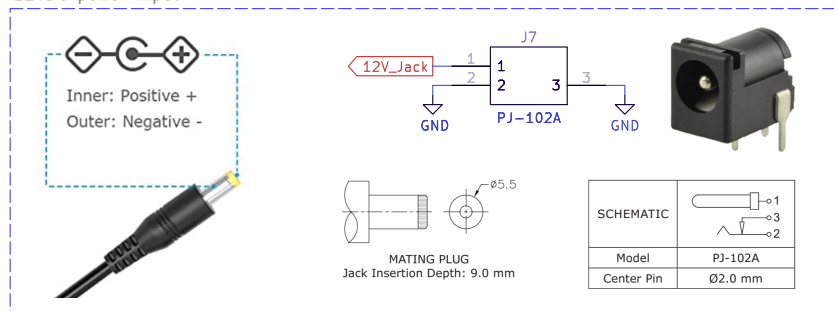
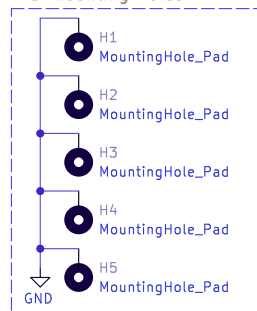
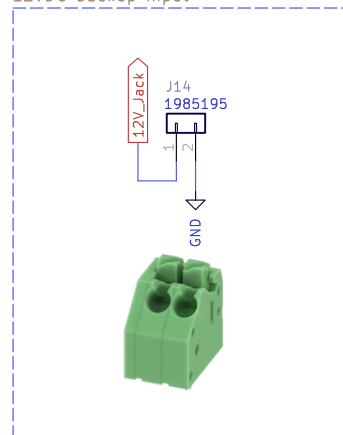
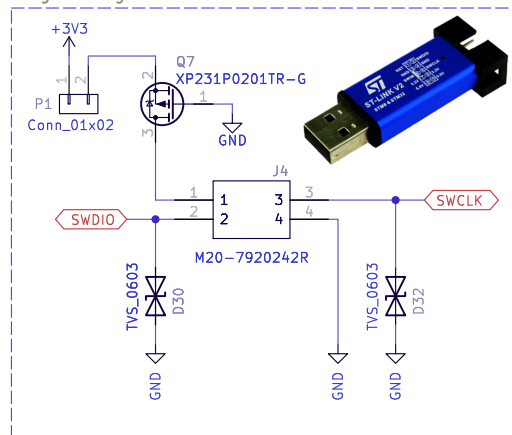
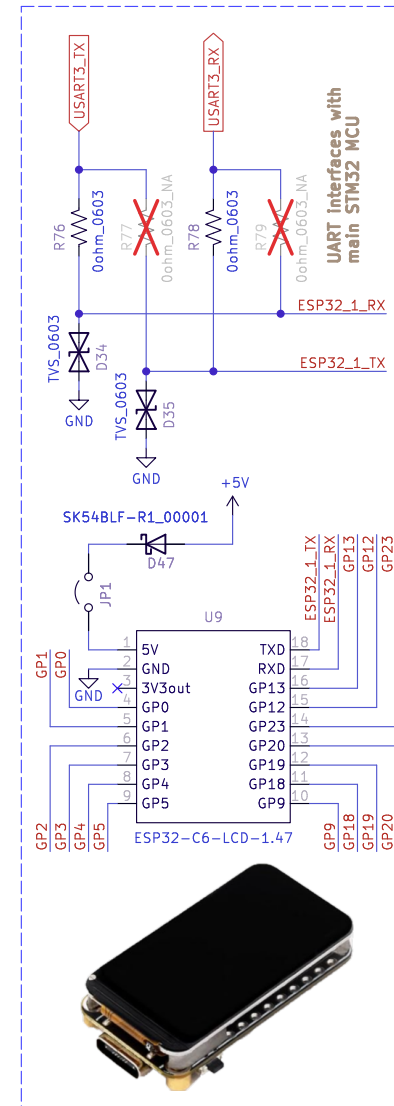
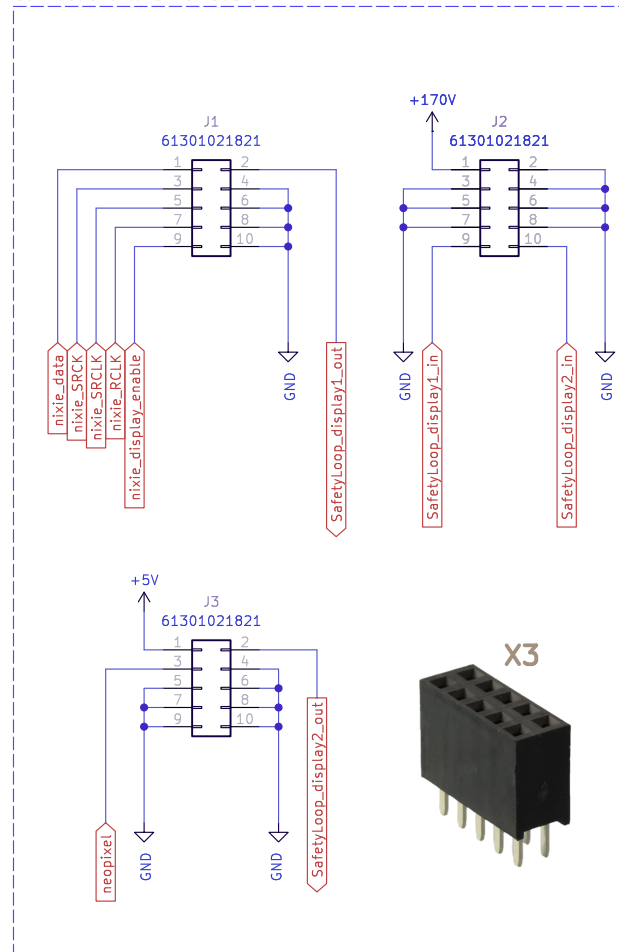
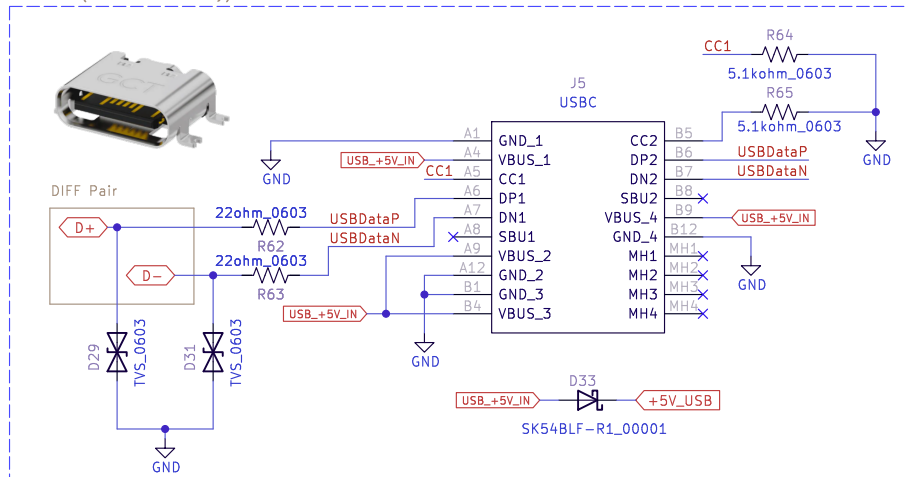
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Date: 2025-12-01

Rev: 1

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Id: 6/9



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Sheet: /Connectors/

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Size: A4

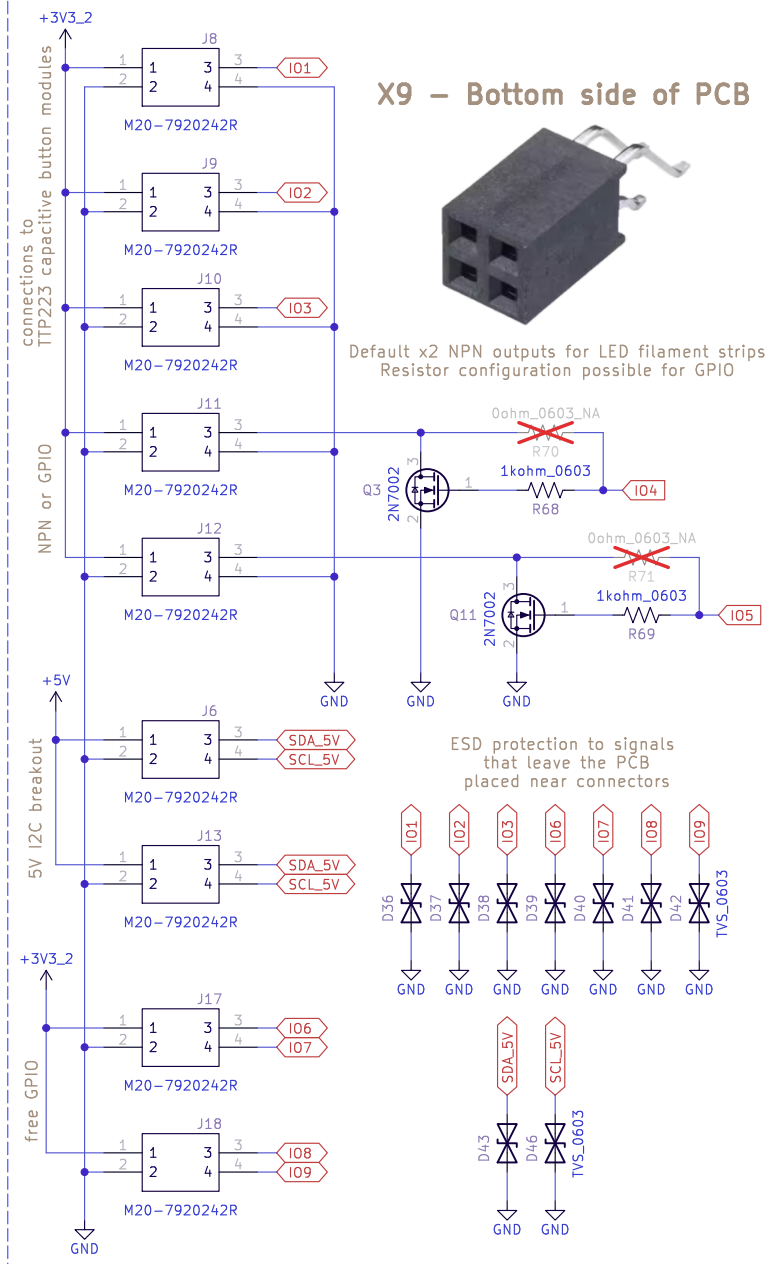
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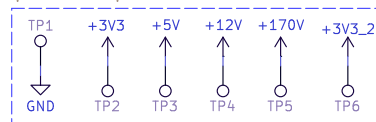
Rev: 1

Id: 8/9

external GPIO optional connections



power test points



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Sheet: /Connectors 2/

File: Connectors 2.kicad_sch

Title: Connectors

Size: A4

Date: 2025-12-01

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Rev: 1

Id: 9/9